

Aromal Suresh

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SUMMARY

Results-driven AI enthusiast with hands-on experience in machine learning, computer vision, NLP, and intelligent automation. Built production-ready systems including YOLOv8-based detection models, RAG pipelines, SEO crawlers, and OCR solutions. Experienced in end-to-end R&D workflows, n8n automation, and AWS deployment.

EDUCATION

Master of Computer Applications (AI & Data Science) <i>JIMS Rohini Sec 3</i>	2022 – 2024
Bachelor of Computer Applications <i>Singhania University</i>	2019 – 2022

PROFESSIONAL EXPERIENCE

Research Analyst <i>Ideas91</i>	Aug 2024 – Present <i>New Delhi, India</i>
– Led AI-focused research and evaluated emerging tools and frameworks for business use cases – Built automation workflows using n8n, reducing manual effort across pipelines – Performed data analysis to support product and technical decision-making – Collaborated with cross-functional teams on AI integration projects	

PROJECTS

VisionRAG <i>AI System</i>	Python, YOLOv8, RAG, FastAPI, ChromaDB
– Architected a secure multimodal AI system integrating YOLOv8 object detection with Retrieval-Augmented Generation over vector databases – Built a REST API pipeline enabling image-based querying, combining visual detections with semantic search (ChromaDB + embeddings) – Implemented prompt injection protection, input validation, and controlled LLM inference for secure AI interactions – Optimized end-to-end workflow with local embeddings (SentenceTransformers) and low-latency inference pipelines	
Web SEO AI <i>AI System</i>	Python, LLMs, YOLOv8, Web Crawling
– Engineered a production-ready AI web crawler performing technical and content SEO audits with automated scoring and issue detection – Integrated YOLOv8-based visual analysis to evaluate UI components (navbar, buttons, layouts) from website screenshots – Built a multi-module pipeline including crawler, SEO analyzer, report generator, and fine-tuning dataset creator (JSONL for LLMs) – Generated structured SEO reports with actionable recommendations, improving website optimization workflows	
LLM Inference Calculator <i>Web Tool</i>	React, TypeScript, AI Systems
– Developed a client-side web application to estimate VRAM, RAM, and storage requirements for running LLMs across multiple architectures and quantizations – Implemented tensor-based memory calculations (weights, KV cache, activations) to model real-world inference constraints – Built a GPU recommendation engine mapping model requirements to hardware (RTX 3060, 4090, A100) with fit classification – Designed dynamic PC/server build suggestions tailored to model-specific compute and memory needs	

SKILLS

Languages: Python, SQL, Java, C
AI/ML: Machine Learning, Deep Learning, Computer Vision, NLP, YOLOv8, RAG, FAISS
Frameworks: FastAPI, Streamlit, n8n
Cloud & Tools: AWS, Git, GitHub
Domains: Data Analytics, Product Research, Automation